

Probability Reasoning Abstract

Probabilistic Topic Modeling for Research Trend Analysis: A Hybrid LDA–LLM Approach

Course: 22AIE301

Abstract

Understanding how research interest evolves within a scientific domain is critical for researchers, funding bodies, and institutions seeking to identify emerging directions and avoid redundant work. Traditional literature review is manual, time-consuming, and does not scale to the thousands of papers published annually. This project proposes a **hybrid probabilistic topic modeling framework** that combines **Latent Dirichlet Allocation (LDA)**, a generative probabilistic graphical model, with **modern LLM-based semantic embeddings** to analyze and visualize thematic trends within a research corpus.

LDA treats each document as a probabilistic mixture of latent topics, and each topic as a probability distribution over words, offering an interpretable generative structure grounded in Bayesian inference — directly reflecting core course concepts of latent variable models and probabilistic inference. However, LDA's reliance on word co-occurrence often produces topics that are statistically valid but semantically incoherent. To address this, the project augments LDA with dense embeddings from a pretrained LLM (e.g., Sentence-BERT or a similar encoder), which capture contextual meaning beyond raw word frequency. These embeddings will be used to (i) improve topic coherence through embedding-guided clustering, and (ii) enable semantic similarity search across papers, so a query paper can be probabilistically mapped to its most relevant existing topics.

The pipeline will be applied to a corpus of research papers on a domain of interest (e.g., AI safety, climate policy, or probabilistic ML itself, to be finalized). The system will output: (a) a set of interpretable latent topics with associated probability distributions over terms, (b) a temporal trend graph showing how topic prevalence shifts across publication years, and (c) a document-to-topic probability assignment enabling users to explore corpus structure interactively.

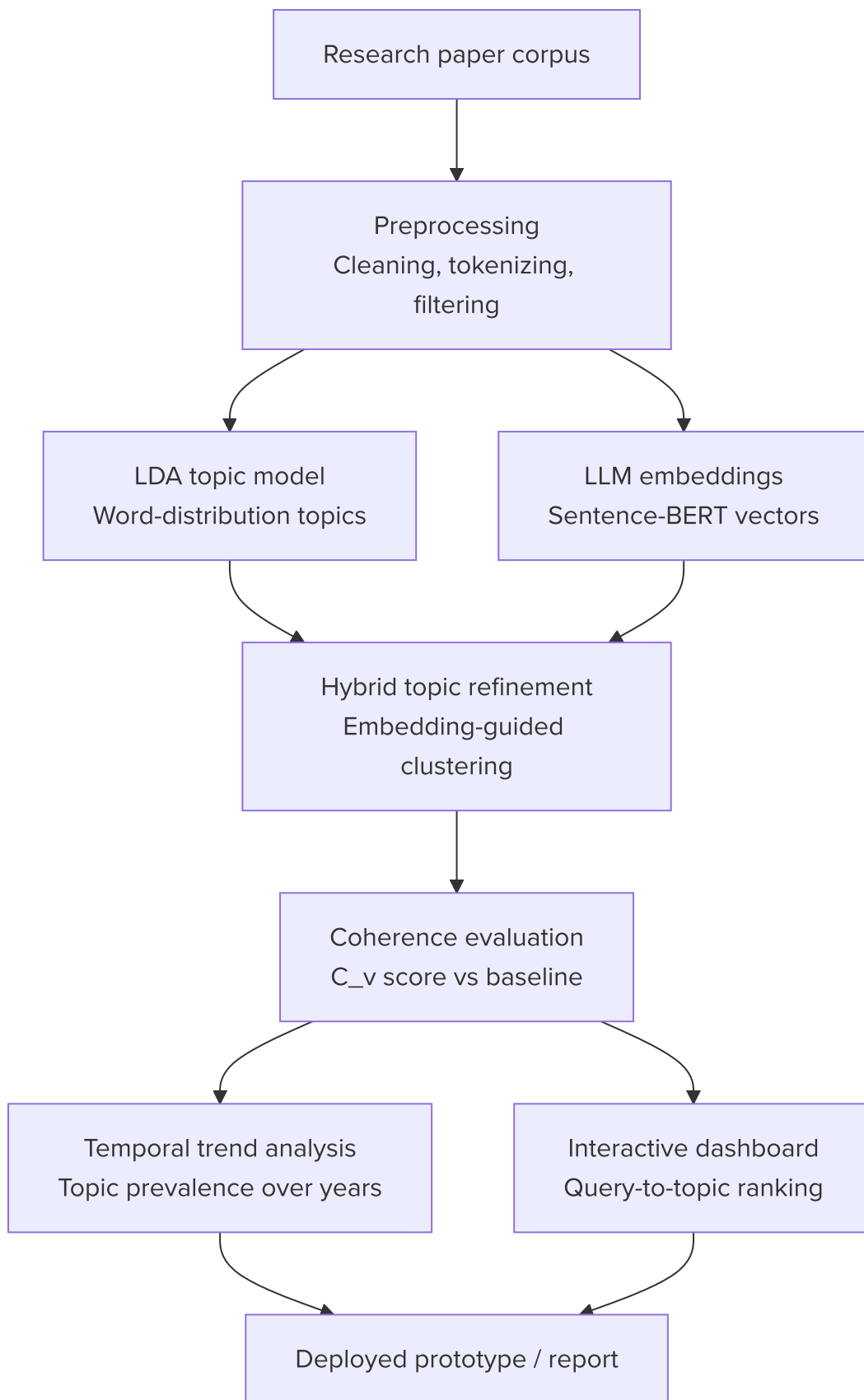
Objectives:

1. Implement a baseline LDA model and evaluate topic coherence using standard metrics (e.g., C_v coherence score).
2. Integrate LLM embeddings to refine and re-rank topics, comparing coherence and interpretability against the baseline.
3. Build a temporal analysis module to visualize how identified topics trend over time.
4. (Stretch goal) Deploy an interactive dashboard/webpage where users can query a topic or paper and receive probabilistically ranked related work.

Expected Outcome: A working prototype demonstrating that combining classical probabilistic generative models with modern LLM embeddings yields more interpretable and coherent topic discovery than either approach alone, with potential extension toward a short paper/report on the comparative evaluation.

Tools/Frameworks: Python, Gensim/pgmpy (LDA), Hugging Face Transformers / Sentence-BERT (embeddings), Scikit-learn, Plotly/Streamlit (visualization).

System Pipeline



1. **Research paper corpus** → raw abstracts/papers collected from the chosen dataset.
2. **Preprocessing** → cleaning, tokenizing, stopword/filter removal.
3. **Parallel modeling paths:**
 - **LDA topic model** → generates word-distribution topics.

- **LLM embeddings** → generates semantic sentence-level vectors (e.g., Sentence-BERT).
4. **Hybrid topic refinement** → embedding-guided clustering merges both signals into sharper, more coherent topics.
 5. **Coherence evaluation** → C_v score compares the hybrid model against the LDA-only baseline.
 6. **Downstream outputs:**
 - **Temporal trend analysis** → topic prevalence tracked across publication years.
 - **Interactive dashboard** → query-to-topic probabilistic ranking for related-work discovery.
 7. **Deployed prototype / report** → final demo and/or short paper.

Candidate Datasets

Dataset	Description	Size	Source
arXiv Metadata Dataset	Titles, abstracts, categories, timestamps for CS/AI/ML papers	2M+ papers	Kaggle (Cornell arXiv dump)
Semantic Scholar Open Research Corpus (S2ORC)	Full-text papers with citation graphs across all fields	80M+ papers (subset by field)	Allen Institute for AI (API/bulk)
ACL Anthology	NLP/computational linguistics papers, well-curated	~80K papers	aclanthology.org
PubMed/PMC OA Subset	Biomedical research abstracts and full text	Millions (subset as needed)	NCBI/NLM
NeurIPS/ICML/ICLR Papers (via OpenReview API)	ML conference papers with reviews/metadata	~10-20K papers	OpenReview API
Custom domain-specific scrape	e.g., AI safety papers via SciSpace/Consensus AI export	Variable	SciSpace, Consensus AI

Recommendation: Start with the **arXiv metadata dataset (cs.AI, cs.LG, cs.CL categories)** — pre-cleaned, includes timestamps essential for trend analysis, and

manageable in size (~50K–100K papers if filtered by category/year).

Use Cases

- 1. Research trend discovery for a subfield** A student/researcher entering a new subfield (e.g., "AI safety") uploads or points to a paper corpus and instantly sees the 8–10 dominant themes, without reading hundreds of abstracts manually.
 - 2. Literature review acceleration** Given a new paper's abstract, the system returns the top-k probabilistically similar existing topics/papers — helping check novelty and avoid duplicated work (directly relevant to this course's own zeroth-review anti-plagiarism goal).
 - 3. Institutional/departmental research mapping** A university department could use this to visualize how its faculty's collective research output has shifted over 5–10 years — useful for strategic hiring or grant positioning.
 - 4. Conference trend tracking** Applied to a specific venue's yearly proceedings (e.g., NeurIPS 2020–2026), it shows how topic popularity rises/falls (e.g., diffusion models emerging, GANs declining).
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Novelties

- 1. Probabilistic backbone with modern refinement** — Most trend-analysis tools either use pure LLM clustering (loses probabilistic interpretability) or pure LDA (loses semantic coherence). This project keeps LDA's generative probability structure as the backbone (aligned with course fundamentals) while using LLM embeddings only to *refine* cluster assignments — not replace the probabilistic model.
- 2. Coherence-driven hybrid weighting** — Rather than naively concatenating LDA and embedding outputs, the project introduces a tunable parameter (α) that blends word-distribution similarity and embedding similarity, then empirically finds the α that maximizes topic coherence — this tuning process itself is a novel mini-contribution.
- 3. Temporal probability drift** — Extending static topic modeling into a **dynamic view**, tracking how the probability mass of each topic shifts year-over-year

(similar in spirit to a Dynamic Bayesian Network's state evolution), rather than just a one-time snapshot.

4. **Anti-duplication application** — Framing the tool specifically as a "check if my project idea has been done before" utility gives it a concrete, immediately useful deployment angle (directly usable by classmates for the zeroth review).

Suggested Evaluation Metrics

- **Topic coherence (C_v, UMass)** — compare baseline LDA vs hybrid model
 - **Topic diversity** — % of unique top-words across topics (avoid redundant topics)
 - **Human judgment / intruder-word test** — sanity check on a sample of topics
 - **Query-to-topic retrieval accuracy** — for the "find related work" use case, manually verify top-5 retrieved papers are genuinely relevant
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